The Shape of Wisdom: How Language Models Converge on an Answer

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Abstract

When a language model answers a factual question, where does its internal state go? We asked four open models (Llama-3.1-8B, Qwen2.5-7B, Gemma-2-9B and Llama-3.2-1B) 2,256 factual questions whose answers recur, such as the country a person is a citizen of. At every layer, we recorded the state from which the answer is produced. Questions that receive the same answer converge on a shared region of the space of states, which we call the answer's *place*. The answer becomes readable from the state suddenly: in every model, readability jumps within a single layer, at a depth between 47% and 71% of the way through the network. A place does not depend on how the question is asked. For example, the place of a country learned from questions about people's citizenship also identifies that country when the question is about a town or a film. Places are also used by the model. After the jump, moving the state of a wrongly answered question into the right answer's place makes the model give the right answer in 83%–97% of cases late in the network, against 2%–4% for a random move of the same size. Before the jump, such moves do nothing. Finally, places reveal which mistakes are near misses. When a model answers wrongly but the right answer's place is close behind the place its state reached, the model usually picks the right answer when the question is asked as multiple choice. This geometry predicts such near misses better than the model's confidence does, in all four models.

1 Introduction

Ask a language model "What country is Aníbal Ponce a citizen of?" and, a few dozen layers later, it answers "Argentina". On the way, the model's internal state at the answering position changes at every layer. This paper asks what shape that change has. Does the state wander, drift slowly toward the answer, or arrive somewhere definite? And if it arrives somewhere, is that the same place for every question with the same answer?

We find that it is. As a question passes through the network, its state moves into a region that it shares with every other question to which the model gives the same answer. We call this region the answer's *place*. Figure 1 shows this in Llama-3.1-8B. Early in the network, questions whose answers are twelve different countries form one tangled cloud. Within a few layers, most of them separate into one strand per country.

We study places from four sides. First, we ask when the answer becomes readable from the state. In each of four models, readability jumps within a single layer, at a depth that differs between models (Section 3.1). Second, we ask whether a place depends on how the question is asked. It does not. The place of a country learned from questions about people also identifies that country in questions about towns and films, and it survives rewording (Section 3.2). Third, we ask whether the model uses places. Moving a state into a place after the jump makes the model give that place's answer. Random moves of the same size rarely do, and moves made before the jump have no effect (Section 3.3).

Fourth, we look at wrong answers. A place is defined by the answer the model gives, not by the true answer, so a wrong answer also lands in a place. Some wrong answers are near misses: shown the options, the model picks the right one. Others are real gaps in knowledge. The model's confidence barely tells the two apart, but where the state sits does. In a near miss, the right answer's place is close behind the place the state reached (Section 3.4).

In these models and on these questions, the answer does not sharpen steadily layer by layer. It appears at one depth, in a region shared with every other question that gets the same answer. A confident mistake reaches the region of the answer it gives in the same way.

2 Methods

2.1 Questions

To test whether questions with the same answer reach the same state, we need many different questions for each answer. We built such a set from Wikidata [36]. It has 2,256 facts and 62 possible answers of four kinds: 24 countries, 16 cities, 12 languages[1] and 10 sports. Each kind of answer is reached through one to three relations, or types of question. A country answers which country a person is a citizen of, which country a town is in, or which country a film is from. A city answers where a person was born or where a company is based. A language answers which language a person spoke or a film was made in. A sport answers which sport an athlete plays.

Models know popular facts far better than rare ones [22], so we sampled subjects evenly from three popularity bands (by number of Wikipedia language editions), giving a mix of

[1]We dropped Hindi as the language a person spoke after a manual check found its Wikidata labels unreliable. Hindi remains as the language of films.

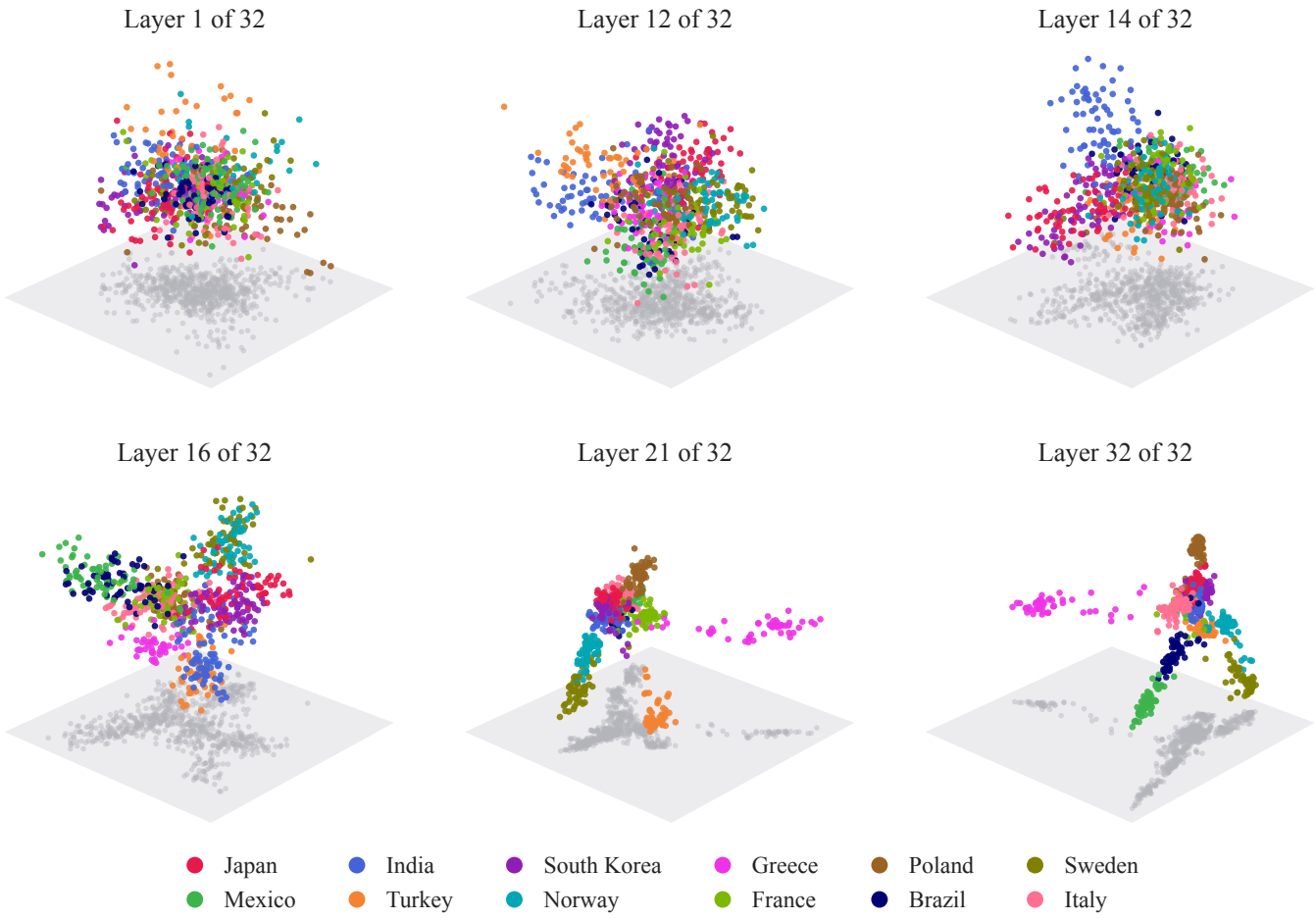


Figure 1: States of questions at six layers of Llama-3.1-8B. Each point is one test question about a person, town or film whose answer is one of twelve countries; colour is the country the model answered. Each panel shows the questions' states at one layer, viewed along the three directions that best separate the twelve countries among the training questions at that layer (found by linear discriminant analysis). At layer 1 all questions share one cloud. At layers 12 and 14 a few countries, such as India, already stand apart. Answers become readable at layer 15. By layer 16 most countries have begun to separate, and at the last layer most form a strand of their own. Each panel is scaled separately; grey dots are shadows on the floor.

facts the models know and do not know. Every subject has one value for its relation, a unique English label and no namesake of the same kind.

2.2 Wordings and formats

Each fact is asked as an open question in up to three wordings. One is a plain template: "What country is Aníbal Ponce a citizen of?" Two are rewordings in assigned styles, such as a casual text or a quiz host's question: "Hi, I'd appreciate knowing which country Aníbal Ponce is a citizen of, if possible?" A language model wrote the rewordings, knowing the subject but never the answer. We kept a rewording only if it named the subject exactly, asked a question, and mentioned no candidate answer or its demonym; 4,323 passed.

The plain question is also asked as four-option multiple choice, in all four cyclic orders of the options. For a random 30% of facts, it is also asked with a wrong suggestion from the user: "In which country is Ireneo Portela? I'm fairly sure it's Mexico." In total, each model sees 16,294 prompts in its own chat format.

2.3 Models and recordings

We study four instruction-tuned models: Llama-3.1-8B-Instruct [12], Qwen2.5-7B-Instruct [32], Gemma-2-9B-it [7] and Llama-3.2-1B-Instruct. All answer greedily.

A transformer builds its output in the residual stream, a vector at each token position that every layer reads and adds to. The answer is produced from the last position of the prompt, so we record the residual stream there before the first layer and after every layer (before the final normalisation). We call this vector the question's *state* at that layer, written h_q^ℓ for question q at layer ℓ. It has one number per dimension of the residual stream, for example 4,096 in Llama-3.1-8B. Depth is the fraction of the model's layers that the state has passed through.

A free-form reply is graded by its first line. If that line names one candidate, matched against hand-curated names such as "China" or "Mandarin", the reply is right or wrong. A single answer outside the candidate list is wrong. Replies that

hedge, refuse or name two answers are set aside as unclear.

2.4 Training and test questions

Several of our measurements first fit something to questions and then test it, for example the answer places defined next. To keep each test fair, fitting and testing use different questions. We split the subjects (the people, towns, films, companies and athletes) at random into two halves, once, and use the same split throughout. Questions about the first half are *training questions*; questions about the second half are *test questions*. Anything fitted uses only training questions, and every number we report is measured on test questions. We split by subject rather than by question because the several questions about one subject, in different wordings and formats, are close copies of each other.

2.5 Answer places

An answer's *place* at a layer is the average state, at that layer, of the training questions to which the model gave that answer:

$$p_a^\ell = \frac{1}{|Q_a|} \sum_{q \in Q_a} h_q^\ell, \tag{1}$$

where Q_a is the set of training questions that the model answered with a. For example, the place of Argentina at layer 15 of Llama-3.1-8B is the average layer-15 state of the training questions that Llama answered with "Argentina", whether about a person, a town or a film, in any wording. We build a place only for an answer the model gave at least five times, for at least three different subjects.

Places are built from the answers the model gives, not from the true answers. A question that the model answers wrongly with "Chile" counts toward the place of Chile. This lets us ask, later, where wrong answers sit.

2.6 Readability

Readability asks whether the answer the model gave can be identified from where its state lies. A state and a place are vectors of the same size, so we can measure the distance between them; no output probabilities are involved. For a test question q, we find the nearest place among answers of the same kind (for a country question, among the country places):

$$\hat{a}(q) = \arg\min_{a \in A_q} \left\| W(h_q^\ell) - W(p_a^\ell) \right\|, \tag{2}$$

where A_q is the set of places of the question's kind and W is a whitening, described below. Readability at a layer is the share of test questions for which $\hat{a}(q)$ is the answer the model gave, counting only questions whose given answer has a place. If states carried no information about the answer, readability would be about one over the number of places, for example 4% for the 24 countries.

Plain distances would mislead, because a few dimensions of the residual stream take very large values and dominate any distance. The whitening W prevents this. At each layer, it re-expresses a state in the 128 directions along which training states vary most, and rescales each direction to unit variance, so that each direction counts equally. To avoid amplifying noise, each variance is first shrunk by 10% toward the average variance.

2.7 Tightness

Readability depends on the whitening. Tightness gives a second view that does not: it measures, in the model's own states, how closely test questions gather around their places compared with how far apart the places are. At each layer,

$$T^\ell = \frac{\frac{1}{N} \sum_q \| h_q^\ell - p_{a(q)}^\ell \|^2}{\frac{1}{|A|} \sum_{a \in A} \| p_a^\ell - \bar{p}^\ell \|^2}. \tag{3}$$

The upper sum runs over the N test questions whose given answer $a(q)$ has a place. A is the set of places of one kind of answer, and $\bar{p}^\ell$ is their average. The numerator is the average squared distance of a question from its own place. The denominator is the average squared distance of a place from the centre of all places. A small T means that questions sit close to their places relative to the distances between places. We compute both parts for each kind of answer and add them, weighting each kind by its number of test questions.

2.8 Uncertainty and controls

Questions about the same subject are not independent: a model tends to get all of them right or all of them wrong. We therefore measure uncertainty by resampling subjects. We draw test subjects at random, with replacement, 300 times, recompute the measure on the questions of the drawn subjects each time, and report the range that holds the middle 95% of the results.

Two controls show how much readability could arise without information about the answer in the states. In the first, we shuffle the given answers between facts of the same kind before building places, and average over five shuffles. This keeps the number and sizes of the places but breaks the link between a state and its answer, so it measures readability by chance. In the second, we replace each state with a representation of the question's text alone: weighted counts of its two- to four-character sequences, reduced to 256 dimensions. This measures how much of the answer can be guessed from the wording, for example from the spelling of a name.

3 Results

3.1 Onset of answer readability

Figure 2A shows readability layer by layer. In every model the curve has the same shape: a low plateau, a jump, and a high plateau. The jump takes a single layer. In Llama-3.1-8B, readability rises from 32% at layer 14 to 71% at layer 15. The other models show the same one-layer jump (Table 1). Readability then keeps rising, to a peak of 86%–94%.

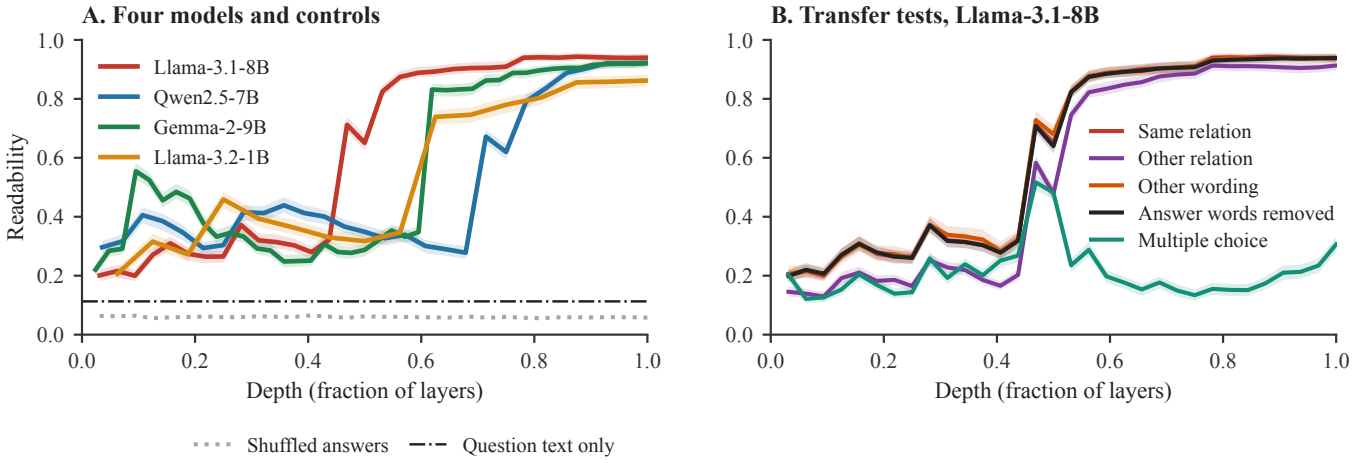


Figure 2: Readability by depth. Readability is the share of test questions whose nearest place is the answer the model gave (Section 2.6). Bands are 95% intervals from resampling subjects. (A) All four models, with two controls: answers shuffled between facts, and the question's characters alone in place of the model's state. (B) Transfer tests in Llama-3.1-8B. "Other relation": places built from one relation, tested on the others. "Other wording": places built from plain questions, tested on rewordings. "Answer words removed": the directions the model uses to write the answer names removed from all states. "Multiple choice": free-form places tested on multiple-choice states. The same-relation, other-wording and answer-words-removed curves nearly coincide.

We call the depth of the jump the *onset depth*. It differs between models, from 47% to 71% of the way through the network (Table 1). To choose it without looking at test questions, we split the training subjects in half again, build places from one half, measure readability on the other, and take the first layer at which readability covers half the distance from chance to its peak.

Before the jump, 20%–55% of answers can already be read, more than with shuffled answers (6%) or from the question's characters alone (11%–18%). This early signal may come from surface cues in the subject's name, such as its language; we did not test this. It does not grow toward the jump.

Tightness. Readability uses whitened states, so the jump could in principle come from the whitening. Tightness uses the model's own states and shows the same event. It falls 16- to 25-fold through the network (Figure 3). The fall is steepest at the onset: the drop from the layer before the onset layer to the layer after it is 16%–32% of the whole fall in log tightness. After that, tightness keeps falling more slowly for about ten layers.

Is this fall a step or a steady contraction? We fitted two curves to log tightness as a function of depth. A straight line means the same contraction at every layer, the pattern found in trained classifiers [17]. A smooth step (a logistic curve) means a sudden change. To compare them fairly, we left out one layer at a time, fitted each curve to the remaining layers, and measured the squared error in predicting the left-out layer. The step predicts better in every model, with 5 to 24 times less error (Table 1). The step is not instantaneous, though: the fitted curve goes from 10% to 90% of its drop over 26% to 44% of the network's depth.

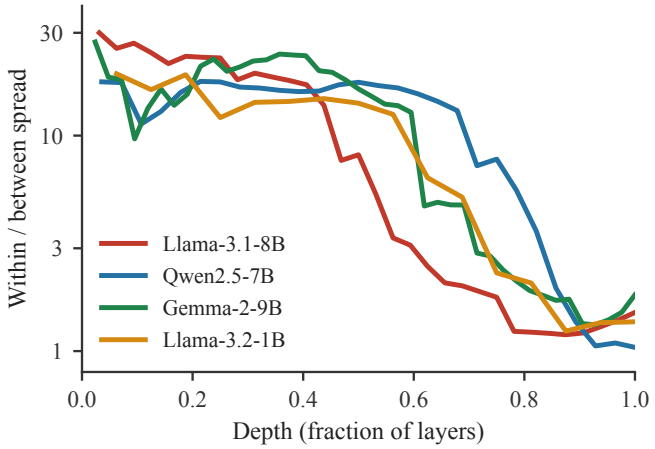


Figure 3: Tightness by depth. Average squared distance of test questions from their own place, divided by the average squared distance of places from their centre, in each model's unwhitened states (log scale). Lower is tighter.

3.2 Transfer across relations, wordings and formats

A place is shared by very different questions. To test this, we built places from the training questions of one relation only, for example people's citizenship, and measured readability on test questions of the other relations with the same kind of answer, for example the country of a town or a film. At 90% of depth this cross-relation readability is 91% in Llama-3.1-8B, against 94% within the same relation (Figure 2B). Likewise, places built from plain questions identify the answers to rewordings just as well (94%). A place therefore does not depend on how the question is asked.

A place could still be just the model's plan to say a particular word, since such a plan would also be shared across questions. One test speaks to this. To produce a word, the

	Llama 8B	Qwen 7B	Gemma 9B	Llama 1B
Layers	32	28	42	16
Accuracy	63%	58%	67%	48%
Onset depth	0.47	0.71	0.62	0.62
Onset layer	15	20	26	10
Readability, layer before	32%	28%	35%	35%
Readability, onset layer	71%	67%	83%	74%
Peak readability	94%	92%	92%	86%
Error, step fit	0.025	0.017	0.051	0.034
Error, line fit	0.137	0.422	0.290	0.198
Other relation	91%	85%	90%	76%
Answer words removed	94%	90%	91%	85%
Wrong answers	339	463	364	488
AUC, confidence	0.69	0.60	0.50	0.61
AUC, geometry	0.75	0.74	0.73	0.83
Gain, 95% interval	0.04–0.13	0.08–0.19	0.15–0.29	0.16–0.32

Table 1: Per-model summary. Accuracy: share of plain questions about test subjects answered correctly (unclear replies count as not correct). Onset depth: the fraction of layers at which readability first covers half the distance from chance to its peak, chosen on training questions. Step and line fits: mean squared error in predicting a left-out layer's log tightness. Other relation and answer words removed: readability at 90% of depth. Bottom block (Section 3.4): wrong answers tested, AUC for predicting recognition, and the 95% interval for the gain in AUC from adding geometry to confidence.

model scores each token by the dot product of its final state with a fixed vector for that token, its unembedding vector, after a normalisation that rescales each dimension. These vectors, rescaled the same way, are the directions through which the state writes words. We took them for every token of the 62 answer names and removed all the directions they span from every state, keeping only the part of each state at right angles to them. Readability did not change (94% at 90% of depth). Removing a random set of directions of the same number did not change it either. Reading a place therefore does not need the directions that write the answer; the place is also present in other directions.

The multiple-choice format shows where sharing stops. Here the model answers with a letter. We asked whether a multiple-choice state lies nearest the free-form place of the option the model chose. Because the two prompt formats differ, all multiple-choice states are shifted relative to free-form states. We estimated this shift as the difference between the two formats' average training states and subtracted it. At the onset depth, the multiple-choice state briefly does lie nearest the chosen option's place: 52% of the time in Llama-3.1-8B, against 6% with shuffled answers. It then falls to 21% as the state turns toward a letter. This fits the finding that models choose the content of an option before binding it to its letter [39].

3.3 Interventions on the state

Readability does not show that the model uses a state's position. To test this, we moved states and checked whether the answer changed. We took test questions that the model had answered wrongly, with an answer that has a place (121–233 questions per model). At one layer, and only at the last prompt position, we replaced the state h with $h + (p_{\text{right}} - p_{\text{given}})$, the state moved by the vector from the given answer's place to the right answer's place. The model then continued its computation and generated its reply as usual. We ran two controls. One moves the state by a vector of the same length in a random direction. The other moves the state toward the place of a third answer, neither given nor right. We repeated this at seven depths between 20% and 90% of the network.

Moves work only after the onset depth (Figure 4). At 20% of depth they correct 0%–1% of wrong answers. After the onset, the later the move, the better it works. At about 90% of depth they correct 97% of wrong answers in Llama-3.1-8B, 94% in Qwen2.5-7B, 97% in Gemma-2-9B and 83% in Llama-3.2-1B. Random moves of the same length produce the right answer in only 2%–4% of cases.

The effect is not specific to the right answer. Moved toward a third answer's place, the model gives that answer in 82%–95% of cases. Questions the model had answered correctly, moved toward another answer's place, switch to that answer in 68%–89% of cases. Some wrong answers have no place of their own, mostly because they name something outside the candidate list. For these, we used the average state of all training questions in place of p_{given}. They are corrected less often (58%–74% at 90% of depth, 172–264 questions per model).

Because places move apart with depth, the moves grow from 3%–5% of the state's length at 20% of depth to 42%–49% at 90%. Late moves are large and close to the output, and may partly write the answer directly. The more telling results are the depth at which moves start to work and the failure of random moves.

3.4 Wrong answers

Because places are built from the answers the model gives, a wrong answer lands in a place too. At the onset depth, 45% of Llama-3.1-8B's wrong answers that name a candidate lie nearest the place of the answer it gave (58% in Qwen2.5-7B). In Llama-3.1-8B, 28% of all wrong answers lie nearest the right answer's place instead.

Some wrong answers are near misses: asked the same question as multiple choice, the model picks the right option. We call these wrong answers *recognised*, and we define the term strictly: the model must pick the right option in at least three of the four option orders. This holds for 6%–60% of wrong answers, fewest in Llama-3.2-1B.

Can these near misses be told apart from real gaps in knowledge? We compared two sets of signals. The first is the model's confidence in its wrong reply: the reply's log-probability and its length. The second is the geometry of the state at the onset layer, measured in whitened states: its distance to the right answer's place, the rank of that place among all places by distance, and how much nearer the nearest place is than the second nearest. For each set, we trained a logistic regression to predict recognition. We scored it by AUC, the probability that a recognised wrong answer gets a higher predicted score than an unrecognised one (0.5 is chance, 1 is perfect). The regression must be scored on wrong answers it was not fitted on, so we used five-fold cross-validation: we

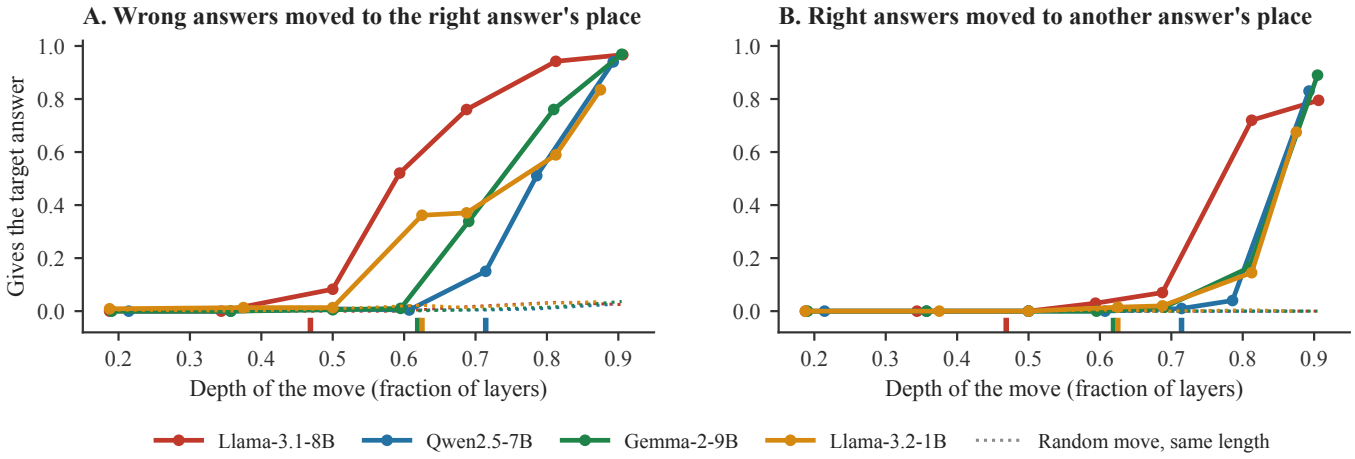


Figure 4: Moving the state toward an answer's place, by the depth of the move. (A) Wrong answers moved toward the right answer's place (solid) or in a random direction of the same length (dotted). (B) Right answers moved toward another answer's place (solid) or in a random direction of the same length (dotted). The y-axis is the share of questions for which the model then gives the answer it was moved toward. Short bars below each axis mark each model's onset depth. Test questions; places from training questions.

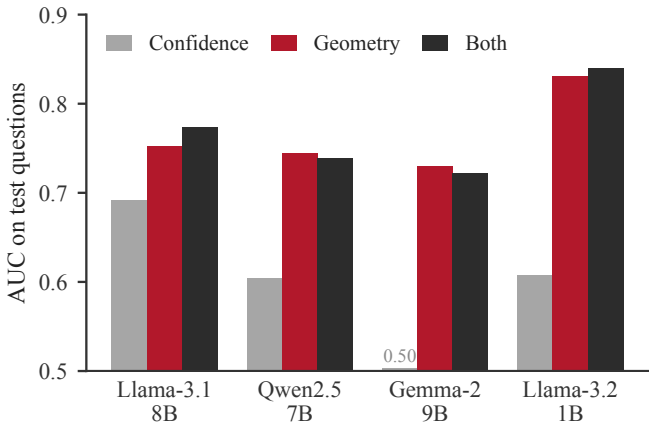


Figure 5: Telling near misses from real gaps. For wrong free-form answers to test questions: how well each set of signals predicts that the model still picks the right option in multiple choice. Confidence: the reply's log-probability and length. Geometry: where the state lies relative to places at the onset depth. AUC on test questions (Section 3.4); 0.5 is chance.

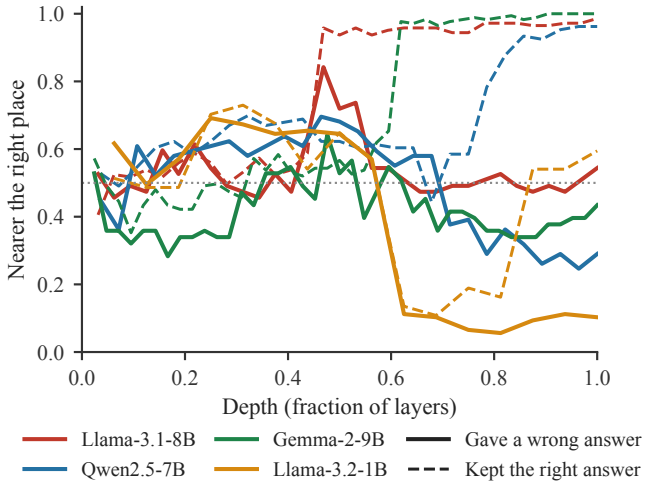


Figure 6: States under a wrong suggestion. Questions that the model answers correctly when asked plainly, asked again with a user's wrong suggestion. Solid: the model then gave a wrong answer. Dashed: it kept the right answer. The y-axis is the share of states that lie nearer the right answer's place than the suggested answer's place.

split the wrong answers into five groups, keeping each subject in a single group, trained on four groups, predicted the fifth, and repeated this for every group.

Geometry predicts recognition better than confidence in every model (Figure 5). In Llama-3.1-8B, the AUC is 0.75 for geometry against 0.69 for confidence. In Gemma-2-9B it is 0.73 against 0.50, so Gemma's confidence carries no signal here at all. Adding geometry to confidence raises AUC in every model, and the 95% interval for the gain, from resampling subjects, excludes zero in each (Table 1).

The rank of the right answer's place shows what the geometry sees. Rank all places by their distance from the state, with 1 for the nearest. For recognised wrong answers, the right answer's place has a median rank of 1–2, depending on the model. For unrecognised ones, its median rank is 4–8. So a near miss is visible inside the model: the right answer's place is among the nearest, often just behind the place of the answer the model gave. The model's confidence in its wrong reply does not show this.

A wrong suggestion sometimes makes a model give a wrong answer to a question it otherwise answers correctly (Figure 6). When the model keeps the right answer, its state goes to the right place and stays there: at the last layer of Llama-3.1-8B, 99% of these states lie nearer the right place than the suggested one. When Llama-3.1-8B gives the wrong answer, its state first moves toward the right place at the onset depth (84% nearer it at layer 15). It then drifts back toward the suggestion (54% at the last layer; 57 questions). The other models also drift away from the right place late in the network, but show no clear early approach to it.

3.5 Given versus true answers

Places follow the answer a model gives, not the true answer. When two models give different answers to the same question, each model's state lies nearer the place of its own answer than the place of the other model's answer (both places from that model) in 66%–88% of cases, across all 12 ordered pairs of models. And places built from the given answers identify those answers far better than places built from the true answers identify the true answers. At 90% of depth, readability is 94% against 75% in Llama-3.1-8B, and 90% against 65% in Qwen2.5-7B.

4 Related work

Reading intermediate layers. The logit lens [26] decodes an intermediate state through the model's output layer, and the tuned lens [1] learns a decoder for each layer. Both show predictions sharpening with depth. Future Lens [28] and Patchscopes [10] read richer content from a single hidden state. All of these read the state through the output vocabulary. Our places are fitted from the states themselves, and reading them does not need the answer words' output directions.

How depth is used. Several studies find that layers play different roles. Early layers gather and detokenise, middle layers build abstract features, and late layers refine a prediction [3, 4, 13, 20, 35]. Correct and incorrect predictions follow similar confidence curves across layers [19]. In factual recall, information about the relation and then the subject moves to the last position, where attention heads extract the attribute [9]. Feed-forward layers act as key–value memories [8], and middle-layer feed-forward modules at the subject's last token are causally decisive [24]. Our onset depth marks, for a whole population of questions, the depth at which each model's answer becomes available at the last position.

Representation geometry. Middle layers represent the same content in a shared space across languages and modalities [37, 41]. Concepts have structured geometry. Categories form simplices [30], some features are circular [5], space and time are linearly decodable [14], counts lie on curved manifolds that attention reshapes [15], and truth has a linear direction [23]. In trained classifiers, representations of different classes separate by a roughly constant factor per layer [17]. Language models move partway toward neural collapse, in which states gather at their class means, at their top layer, and further with scale [40]. The residual stream contains stable regions, within which small changes leave the next-token prediction unchanged [18]. Yet, almost surely, no two prompts map to the same last-position state [25]. Places fit both findings: questions can gather in the directions that carry the answer while staying distinct in others.

What models know but do not say. Internal states often encode more than the output shows [6, 27, 33]. On knowledge benchmarks such as MMLU and ARC, however, this gap is near zero [31], and internal states mostly reflect what the model recalls rather than what is true [2]. False few-shot demonstrations can override a correct internal answer at a critical layer [16], and users' suggestions shift answers [34]. In multiple choice, attention in a few middle layers selects the answer letter, and a sparse set of heads then raises its probability [38]. The winning option is chosen by its content before it is bound to its letter [39], and models carry a prior over letters [42].

Steering. Adding a direction to the residual stream changes a model's behaviour [21, 29]. Our moves are of this kind. Each one, however, takes a single question's state from one measured place to another at one layer, and is compared with random moves of the same length.

5 Limitations

The questions are short and factual, and their answers come from a closed set of 62 countries, cities, languages and sports. Three of these four kinds are names. For the one kind that is not, sports, the rise at the onset layer is smaller (7–19 percentage points, against 41–52 for names), and readability at 90% of depth is lower (63%–90% against 89%–94%). Sports have at most ten places, all from one relation, so we cannot tell whether this reflects the kind of word or the smaller set. Places for open-ended answers, multi-step reasoning or long contexts may look different.

We studied four instruction-tuned models between 1 and 9 billion parameters, with one prompt format each and greedy answers. Readability and tightness both measure where states sit, not the computation that moves them there. The step fit shows that a sudden change describes tightness much better than a constant contraction; it does not show that tightness changes within a single layer. Moves at late layers may partly write the answer directly, so the strongest causal evidence is the depth at which moves start to work and the failure of random moves.

Wikidata answers are a reference, not ground truth. We removed one unreliable relation–answer pair after a manual check, and a small number of wrong labels surely remain. The rewordings were written by a language model and filtered automatically.

6 Conclusion

We followed the state from which four language models answer short factual questions, layer by layer. In all four, the answer becomes readable from the state within a single layer. The depth of this onset differs between models. From the onset on, questions with the same answer gather in a shared place. The place does not depend on the relation or the wording of the question, and it can be read without the directions that write the answer's name. Moving a state into a place after the onset makes the model give that place's answer; before the onset, the same move does nothing. Places follow the answer the

model gives rather than the true one, so wrong answers land in places too. Where they land separates near misses, which the model recognises when shown the options, from real gaps in knowledge, and it does so better than the model's confidence. Whether the same picture holds for open-ended answers, for reasoning over several steps and for larger models is the natural next question.

Code and data. Code, the dataset and all derived results: https://github.com/gut-puncture/The-Shape-of-Wisdom.

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